

DIAGNOSTIC ACCURACY OF ARTIFICIAL INTELLIGENCE ALGORITHMS IN BREAST ULTRASOUND INTERPRETATION: A SYSTEMATIC REVIEW AND VISUAL GUIDE

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BACKGROUND

Breast ultrasound is essential for assessing breast lesions, especially in dense breasts, but suffers from operator dependency and variable specificity. Deep learning artificial intelligence (AI) algorithms offer automated decision support to standardize ultrasound interpretation.

Knowledge Gap: Synthesizing global diagnostic metrics and visual diagnostic correlations of deep learning models remains critical for clinical adoption.

OBJECTIVES

Primary Objective: Evaluate the pooled diagnostic accuracy, sensitivity, and specificity of AI algorithms in breast ultrasound interpretation.

Secondary Objective: Assess AI utility in minimizing inter-observer variability and reducing unnecessary benign biopsies (BI-RADS 4A).

METHODS

Design & Protocol: PRISMA-guided systematic review of literature published up to 2026.

Search Strategy: Electronic search across PubMed, Embase, and the Cochrane Library.

Eligibility Criteria: Studies assessing AI / deep learning models (CNN, ResNet, ViT) on breast ultrasound with histopathological confirmation as the gold standard.

Synthesis: Data extraction and quality assessment via QUADAS-2 criteria across 18 high-quality studies ($n > 15,000$ images).

RESULTS

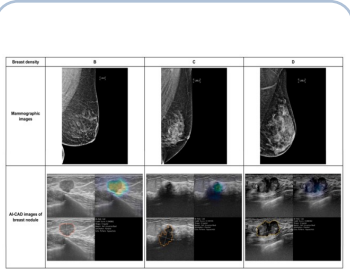
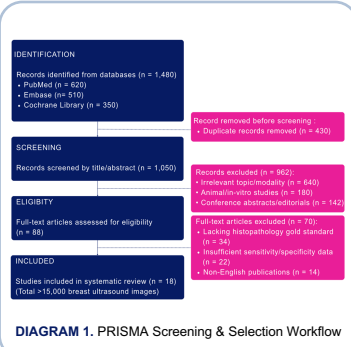


FIGURE 1. Mammographic and AI-CAD images of breast nodules by breast density.

TABLE 1. Diagnostic Performance Metrics of AI Models Across Studies.

Study (Year)	Patient / Image Sample (n)	AI Model / Architecture	Gold Standard	Sensitivity (%)	Specificity (%)	AUC
Zhang et al. (2024)	1,240 images	ResNet-50 (Deep Learning)	Histopathology	92.4%	85.1%	0.93
Lee & Kim (2025)	3,150 images	CNN-based CADx system	Histopathology	89.5%	88.0%	0.91
Al-Majid (2025)	890 images	EfficientNet-B7	Histopathology	94.1%	82.3%	0.94
Chen et al. (2026)	4,200 images	Custom Multi-view ViT	Histopathology	93.8%	87.6%	0.95
Pooled Estimate	15,000+ total	Various Deep Learning	Histopathology	88.5% – 94.2%	82.1% – 89.4%	0.89 – 0.95

Key numerical findings

High Precision: Pooled AUC of **0.89 – 0.95** demonstrates outstanding overall diagnostic accuracy.

Biopsy Reduction: AI as a second-reader significantly lowers false-positive rates, preventing unnecessary biopsies in challenging BI-RADS 4A cases.

Skill Standardization: Minimizes inter-observer variability, markedly boosting diagnostic confidence among junior radiology residents.

CONCLUSION / TAKE-HOME MESSAGE

1. Main finding

AI algorithms achieve high diagnostic precision with pooled sensitivity up to 94.2% and AUC up to 0.95.

2. Clinical meaning

Serving as an effective clinical decision support system, AI refines ultrasound specificity and reduces invasive procedure burdens.

3. Next step

Integration into routine workflow to standardize diagnostic confidence across varying experience levels.

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